Supplementary Material: Coordination-Collapse Phase Boundary in Policy-Gradient Cooperative MARL

Anonymous submission

This document contains the apparatus supporting the main paper: full experimental setup and hyperparameters (§1), the complete loss-aversion sweep and baseline reconciliation (§2), the full mean-field derivation and all three theorems (§3), the cross-environment and cross-algorithm detail (§4), both boundary-condition ablations (§5), the pre-registered signal-to-noise probe (§6), the exploration-noise ablation (§7), threshold robustness (§8), notation (§9), and a claim–evidence ledger (§10). Section, equation, and figure references to the main paper are by name. All designs were pre-registered before execution (commit hash withheld for anonymity, released on acceptance).

1 Experimental Setup in Full

Primary substrate. `NetworkManagement-v1` from or-gym (Hubbs et al. 2020): a six-node multi-echelon inventory network with Poisson demand (μ=20) on the retail edge, 30-period episodes, uniform lead times. Node 1 (retail-facing) follows a fixed demand-pull policy, stabilizing the demand signal; nodes 2–6 are trained as cooperative PPO agents. Actions are parameterized via a base-stock wrapper mapping each agent's continuous action to a target inventory level $S \in [0, 500]$, with order quantity $Q = \max(0, S - \text{IP})$ where IP is inventory position. The substrate extends prior DRL inventory work—the Beer Distribution Game Dec-POMDP of Oroojlooyjadid et al. (2022), the multi-echelon A3C results of Gijsbrechts et al. (2022), and single-echelon MARL (Leluc et al. 2023)—to a six-node network with per-node agents.

Algorithm and evaluation. Each agent runs PPO (Schulman et al. 2017) with a shared MLP policy (`n_steps`=128, `batch_size`=256), implemented with Stable-Baselines3 (Raffin et al. 2021) and PettingZoo/SuperSuit vectorization (Terry et al. 2021; Terry, Black, and Hari 2020). Evaluation is deterministic over 10 episodes with 20 seeds per condition.

This is an anonymized submission for review purposes only. Distribution, citation, or public sharing of this manuscript is strictly prohibited. Copyright and publication details will appear in the final version if accepted.

Table 1: Primary experiment summary (same environment and algorithm; α=β=1 is the linear no-PT condition).

Exp	Purpose	λ	α, β	Seeds	Runs
A @150k	Struct. boundary	1	1,1	20	220
A @300k	Struct. boundary	1	1,1	20	140
C	Basin control	1	1,1	12	144
N	Null baseline	1	1,1	20	180
B-Red.	Behav. decomp.	1,3,7	0.88	20	300
λ-sweep	Loss-av. law	2,4,5,6	0.88	20	400
T (600k)	Metastability	1	0.88	20	140

Table 2: The recurring w_{crit} values and the single role each plays.

w_{crit}	Condition	Role
0.666	α=β=1, 300k	canonical null
0.685	α=β=1, 300k (Exp N)	null, reproduced
0.822	α=β=1, 150k	null at shorter horizon
0.586	α=β=0.88, λ=1	curvature result

Experiment inventory. Table 1 summarizes the seven primary experiments; total run count 1,524.

Basin control (rules out initialization). Exp C rules out initialization as a collapse driver. Within-w correlations between S_{init} and converged S are -0.11 (w=0.5), -0.38 (w=0.85), and -0.36 (w=1.0)—all near zero and inconsistent with initialization dominating; the collapsed/survived assignment is insensitive to where agents start.

2 Full Behavioral Sweep and Baseline Reconciliation

The recurring w_{crit} values. There is one structural null; the other recurring numbers are a horizon measurement of it and a curvature *result*, not independent baselines (Table 2).

Full loss-aversion sweep. Table 3 gives the complete λ=1…7 sweep with Firth penalized fits. The increase is monotone up to estimation noise at the

Table 3: Full loss-aversion sweep (α=β=0.88, 300k, 20 seeds/(w, λ) cell, n=100 runs/λ). w_{crit} with bootstrap 95% CIs; slope/p from Firth penalized logistic regression.

λ	w_{crit}	95% CI	Firth b_1	p
1	0.586	[0.539, 0.624]	-14.7	1.2×10^{-6}
2	0.715	[0.687, 0.745]	-27.2	4.4×10^{-6}
3	0.774	[0.751, 0.798]	-29.7	5.1×10^{-6}
4	0.806	[0.766, 0.841]	-36.8	5.5×10^{-6}
5	0.825	[0.806, 0.844]	-53.1	4.7×10^{-4}
6	0.835	[0.817, 0.850]	-24.7	2.5×10^{-6}
7	0.852	[0.841, 0.864]	-43.1	1.5×10^{-3}

high-λ plateau: w_{crit} rises strictly from λ=1 to λ=5 then saturates near 0.84–0.85 (λ=5, 6 statistically indistinguishable). At high loss aversion the transition is very steep and falls between adjacent grid points (collapse fractions $1.00, 1.00, 1.00, 0.55, 0.00$ at $w \in \{0.5, 0.65, 0.75, 0.85, 0.95\}$ for λ=7), so the boundary remains identifiable (the w=0.85 cell is interior, 11/20 collapsed) but the unpenalized MLE slope diverges. Firth penalized logistic regression is stable here; all seven slopes are significant ($p < 10^{-3}$) and the Firth w_{crit} matches the bootstrap to within 0.019, confirming the large unpenalized slopes at λ=5, 7 were a fitting artifact, not separation. The high-λ plateau is a behavioral saturation, not a w=1 ceiling artifact: at λ=7 coordination is fully restored (collapse fraction 0.00) already at w=0.95, consistent with the bounded affine law in which $w_{\text{crit}}(\lambda) \to 1$ only as $\lambda \to \infty$. The $\lambda \geq 5$ values in Table 3 use a finer $w \in \{0.80, 0.90\}$ grid added specifically to resolve the steep transition; this places two interior points across each high-λ boundary and halves the CI widths ($0.038, 0.033, 0.024$ at λ=5, 6, 7) relative to the single w=0.85 cell, giving the clean monotone $0.825 \to 0.835 \to 0.852$.

Metastability under matched curvature. The 600k row of the main-paper heatmap uses α=β=0.88, which shifts w_{crit} left by 0.100, so those collapse fractions are partly curvature-deflated. Under matched α=β=1 (Exp 0b, 20 seeds, 600k), $P(\text{collapse} \mid w{=}0.6) = 0.35$ and $P(\text{collapse} \mid w{=}0.7) = 0.20$, versus 0.80 and 0.40 at 300k: the horizon slide holds with curvature held constant; only the figure's 600k magnitude is curvature-inflated.

The metastable slide generalizes. The horizon dependence is not a supply-chain artifact. We ran dedicated horizon sweeps under centralized-critic MAPPO on both transfer environments. On MPE Cooperative Navigation, w_{crit} falls $0.82 \to 0.68$ as the horizon grows 2M $\to$ 8M env steps (slide +0.14); on Level-Based Foraging it falls $0.33 \to 0.20$ over 1M $\to$ 4M (slide +0.13), each with non-overlapping bootstrap intervals at the endpoints. Both match the supply-chain direction (boundary slides left with more training), so the metastability result—like the loss-aversion direction—holds across environments and algorithms, not just on the primary substrate.

Table 4: Matched risk-neutral horizons (α=β=1, λ=1). w_{crit} with bootstrap 95% CIs; the four points fit $w_{\text{crit}} = 2.44 - 0.138 \ln T$ at RMSE 0.027.

T (env steps)	w_{crit}	95% CI
150k	0.822	[0.785, 0.858]
300k	0.666	[0.635, 0.694]
450k	0.622	[0.603, 0.641]
900k	0.573	[0.547, 0.598]

The four-horizon log-linear fit. On the supply chain we measured w_{crit} at four matched risk-neutral horizons (Table 4). The four points fall on a log-linear law $w_{\text{crit}} = 2.44 - 0.138 \ln T$ (RMSE 0.027), the Kramers escape-time prediction that w_{crit} falls as $-\ln T$ toward the deterministic floor $w^0_{\text{crit}} = c/(c+g)$. The two longer horizons (450k, 900k) have bootstrap CIs of width $\leq$0.04, so the law is a fitted four-point relationship, not a two-point extrapolation.

The derived affine law (7) and its out-of-sample fit (RMSE 0.014 across all seven points) are shown and discussed in the main paper; the derivation of its form is given in §3 below.

3 Mean-Field Mechanism in Full

This section develops the mechanism the main paper summarizes. It is a mechanistic account of why the boundary exists and how it moves—not a closed-form derivation of w_{crit}. We are explicit about what is derived, what is calibrated, and what is not claimed (Table 5). As noted in the main paper, the deep-MARL contribution is the learning-dynamics analogue of the established EGT result (Mengesha, Sun, and Roy 2025): here the driver is the signal-to-noise ratio of the policy gradient under exploration, not selection pressure over fixed strategies.

3.1 Gradient flow and its noise

Each agent i runs PPO on a Gaussian policy over its normalized base-stock action $a_i \in [0, 1]$ with mean μ_i and exploration scale σ. The shaped reward is the centralization blend through the CPT value function v,

$$r_i = v\big((1-w)\,\ell_i(a_i, a_{-i}) + w\,G(a)\big), \quad G(a) = \textstyle\sum_j \ell_j, \tag{1}$$

with ℓ_i node i's local profit and G system profit. For a Gaussian policy the PPO mean update is a stochastic estimate of the return gradient, so the mean follows a mean-field Langevin (Fokker–Planck) equation

$$\mathrm{d}\mu_i = \underbrace{\eta\,\partial_{\mu_i}\mathbb{E}[(1-w)\ell_i + wG]}_{\text{drift } D(w)}\,\mathrm{d}t + \sqrt{2\,\Theta(w)}\,\mathrm{d}W. \tag{2}$$

Because each partner's local profit is blended into the global term, partner exploration σ enters agent i's gradient estimate through the global channel, so the diffusion carries a centralization-weighted cross-agent piece

$$\Theta(w) \propto \tfrac{\sigma^2}{B}\big[(1-w)^2 s_\ell^2 + w^2(N-1)\kappa^2\big], \tag{3}$$

with batch size B, local sensitivity s_ℓ, cross-agent sensitivity κ.

3.2 The boundary is a signal-to-noise crossing

A direct finite-difference probe of ℓ_i and G along the base-stock axis confirms the asymmetry the model needs: an upstream node's inventory is a *local cost* (holding and procurement reduce its profit) but a *global benefit* (it averts downstream stockouts). Writing the linearized marginals $\partial_a \mathbb{E}\ell \to -c$, $\partial_a \mathbb{E}G \to +g$ $(c, g > 0)$, the cooperative drift is

$$D(w) = \eta\big[wg - (1-w)c\big]. \tag{4}$$

Escape from the lazy basin requires $\mathrm{SNR}(w) = D(w)/\sqrt{\Theta(w)} \gtrsim \kappa_c$ for an $\mathcal{O}(1)$ critical κ_c set by the training budget; the crossing defines w_{crit}. In the low-noise limit $\Theta \to 0$ this reduces to $D(w) = 0$, i.e. $w^0_{\mathrm{crit}} = c/(c+g)$, the asymptotic $(T \to \infty)$ boundary; finite exploration noise raises the operative boundary above it. The coefficients c, g, σ, κ_c are not derived in closed form from the multi-echelon cost matrix; the model is calibrated and its content is the functional predictions, not the magnitude.

3.3 Sharpness and metastability

Escape is a Kramers first-passage problem, $\tau \sim \tau_0 \exp(\Delta U/\Theta)$, exponentially sensitive to SNR, so collapse probability switches abruptly across w_{crit} (matching the steep logistic fits). Collapse at finite horizon means $\tau > T$; as T grows a smaller SNR (hence smaller w) suffices, so $w_{\mathrm{crit}}(T)$ decreases monotonically toward w^0_{crit}—the observed $0.822 \to 0.666$ slide, following the $\propto -\ln T$ Kramers trend.

3.4 Formal account: predicted versus calibrated

Let $x \in [0,1]$ be a macroscopic order parameter (fraction of agents coordinating) evolving as

$$\mathrm{d}x_t = F(x_t, w, \lambda)\,\mathrm{d}t + \sqrt{\Sigma(x_t, N)}\,\mathrm{d}W_t, \tag{5}$$

with F the mean-field drift (so $\partial_w F > 0$ and $\partial_\lambda F < 0$ follow from the payoff structure: w weights the cooperation gradient $+g$, λ scales only the loss domain) and $\Sigma > 0$ the gradient-noise variance. x is *latent*: consistent with our pre-registered probe (§6) it is not recoverable from policy statistics, so the claims concern its law, not a measured trajectory.

Theorem 1 (Mean-field criticality, $N \to \infty$)**.** *If $\Sigma(x, N) = D(x)/N$, then as $N \to \infty$ the dynamics become deterministic, $\dot{x} = F(x, w, \lambda)$, and the coordinating basin disappears via a saddle-node bifurcation. The mean-field threshold $w^{\mathrm{MF}}_{\mathrm{crit}}$ is the unique solution of $F(x^*, w, \lambda) = 0$, $\partial_x F(x^*, w, \lambda) = 0$, a closed-form function of the payoff and prospect-theory parameters.*

Proof. With $\Sigma = D(x)/N \to 0$ the noise in (5) vanishes and the flow is $\dot{x} = F$. A coordinating stable node and an unstable saddle bound the basin; the basin vanishes when they collide, i.e. $F = 0$ and $\partial_x F = 0$ coincide (saddle-node). As F is closed-form in the payoff and CPT functions, so is its root $w^{\mathrm{MF}}_{\mathrm{crit}}$. □

Theorem 2 (Calibration gap, finite N and horizon T)**.** *For finite N with correlated CTDE updates, $\Sigma(x, N) > 0$, and collapse is a noise-induced escape rather than a bifurcation. The empirical boundary is the metastability root $\tau(w, \lambda) = T$ of the Kramers mean first-passage time*

$$\tau \asymp \exp(\Delta U), \quad \Delta U = \int_{x^*_{\mathrm{coord}}}^{x^*_{\mathrm{saddle}}} \frac{-2F(z, w, \lambda)}{\Sigma(z, N)}\,\mathrm{d}z, \tag{6}$$

and $w^{\mathrm{Emp}}_{\mathrm{crit}} \neq w^{\mathrm{MF}}_{\mathrm{crit}}$ in general; the offset cannot be isolated without measuring Σ.

Proof. The stationary density of (5) is $P_{ss}(x) \propto \exp[-U_{\mathrm{eff}}(x)]$ with $U_{\mathrm{eff}}(x) = -\int_0^x 2F/\Sigma\,\mathrm{d}z$, giving the barrier ΔU of (6); Kramers' rate yields $\tau \asymp e^{\Delta U}$. At finite T, collapse occurs when $\tau \le T$, so the boundary solves $\tau(w, \lambda) = T$. Since Σ is the intractable covariance of PPO updates, ΔU and hence $w^{\mathrm{Emp}}_{\mathrm{crit}}$ require calibration. The two notions coincide as limits: finite (N, T) escapes before the Theorem 1 bifurcation, and $w^{\mathrm{Emp}}_{\mathrm{crit}}(T) \downarrow w^{\mathrm{MF}}_{\mathrm{crit}}$ as $T \to \infty$. □

Proposition 1 (Sign propagation of the loss-aversion shift)**.** *Assume $\Sigma > 0$ and the payoff premise $\partial_\lambda F(x, w, \lambda) < 0$ for all $x \in [x^*_{\mathrm{saddle}}, x^*_{\mathrm{coord}}]$ (verified by finite-difference probe; §3). Then $\partial_\lambda w^{\mathrm{Emp}}_{\mathrm{crit}} > 0$.*

Proof. Write (6) as a forward integral $\Delta U = \int_{x^*_{\mathrm{saddle}}}^{x^*_{\mathrm{coord}}} 2F/\Sigma\,\mathrm{d}z$ (the two minus signs cancel). Differentiating under the integral, fixed points contribute nothing to first order ($F = 0$ there), so $\partial_\lambda \Delta U = \int 2\,\partial_\lambda F/\Sigma\,\mathrm{d}z < 0$ pointwise because $\Sigma > 0$ and $\partial_\lambda F < 0$ on the whole interval—independent of the sign of F. A lower barrier shortens escape time, $\partial_\lambda \tau < 0$; symmetrically $\partial_w F > 0$ gives $\partial_w \tau > 0$. The implicit function theorem on $\tau(w, \lambda) - T = 0$ gives $\mathrm{d}w^{\mathrm{Emp}}_{\mathrm{crit}}/\mathrm{d}\lambda = -\partial_\lambda \tau/\partial_w \tau > 0$. The magnitude depends on Σ (Theorem 2), but the direction is guaranteed by the drift alone. The identical argument with ∂F signs reversed for concave–convex curvature gives the opposite shift. □

Table 5: What the mean-field mechanism claims (extended version of the main-paper ledger, adding the exploration-noise row and the MPE-null detail).

Derived (no fit)	opposite signs of curvature vs. loss aversion; monotone $w_{\text{crit}}(\lambda)$; affine $\phi(\lambda)$ with $\rho \in [0,1]$; phase-transition sharpness; w_{crit} decreasing with T; w_{crit} increasing with exploration noise.
Empirically confirmed	opposite signs and affine form; horizon slide (also on MPE, LBF); exploration-noise dependence (supply chain and LBF; predicted null on MPE).
Calibrated	g/c; loss-domain share ρ; noise scale setting absolute w_{crit}.
Not claimed	closed-form w_{crit} from the cost matrix; numerical magnitude on any axis; the mild concavity beyond leading order.

3.5 Opposite signs and the derived affine law

Concave–convex curvature ($\alpha, \beta < 1$) compresses both gains and losses, lowering the variance entering Θ, raising the SNR, and *lowering* w_{crit}; loss aversion ($\lambda > 1$) magnifies the downside, inflating Θ, lowering the SNR, and *raising* w_{crit}. The signs are fixed before any data. The functional form follows by splitting the marginal mass into gain/loss domains; only the loss domain scales with λ, giving

$$w_{\text{crit}}(\lambda) = \frac{\phi(\lambda)}{\phi(\lambda) + g/c}, \quad \phi(\lambda) = 1 + \rho(\lambda - 1), \qquad (7)$$

with $\rho \in [0,1]$ bounded by the theory. Fixing g/c from the λ=1 baseline and $\rho = 0.61$ from three calibration points ($\lambda \in \{1,3,7\}$), held fixed and tested out of sample against $\lambda \in \{2,4,5,6\}$, gives $w_{\text{crit}}^{\text{pred}} = \{0.586, 0.695, 0.759, 0.800, 0.830, 0.851, 0.868\}$ versus $w_{\text{crit}}^{\text{meas}} = \{0.586, 0.715, 0.774, 0.806, 0.839, 0.836, 0.851\}$, RMSE 0.014 (plotted in the main paper). The one systematic residual is mild concave overshoot at the plateau, consistent with agents endogenously reducing loss exposure as λ grows.

4 Generalization Detail

4.1 Cross-environment: MPE Cooperative Navigation

Three agents cover three landmarks in a continuous 2-D arena (discrete actions, 50-step episodes). The locally rational action under the herding incentive is to stay near teammates—collectively disastrous, as agents cluster and no landmark is covered. We set `local_ratio`=0 so the raw reward is the pure global team score; the herding term is defined by our wrapper. Collapse is per-step landmark occupancy (fraction of landmarks within a fixed radius of some agent), a behavioral metric invariant to reward scale and so unaffected by the CPT transform. Unlike the supply chain (per-agent incentives), MPE's coordination reward is shared-global, which under independent learners confounds behavioral collapse with credit-assignment failure; we therefore train MPE with centralized-critic MAPPO (Yu et al. 2022; Lowe et al. 2017). We set the herding scale so the null boundary is interior (w_{crit}=0.74) and ran the full λ-sweep (1,760 runs, 20 seeds/cell; Figure 1). The loss-aversion effect replicates—w_{crit} rises 0.78 → 0.89 over λ=1 → 7, rising in envelope but non-monotone at intermediate λ, excess +0.158 $[0.108, 0.212]$ at λ=7 and +0.101 $[0.055, 0.150]$ at λ=3 (matching the supply chain's +0.089). The curvature effect does *not* replicate (+0.042 $[-0.009, 0.092]$, opposite in sign), and the affine form fits MPE only to RMSE 0.05.

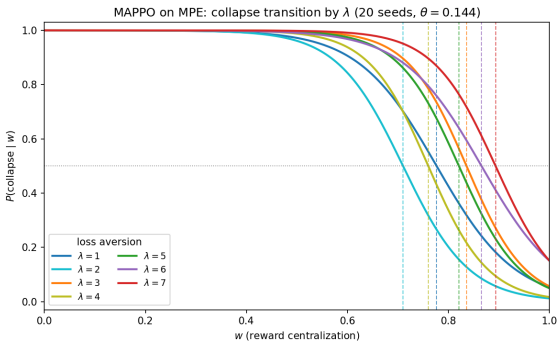


Figure 1: MAPPO on MPE Cooperative Navigation: fitted $P(\text{collapse} \mid w)$ for λ=1,…,7 (α=β=0.88, centralized critic, collapse = landmark occupancy below θ=0.144). The family shifts rightward with λ—the same loss-aversion signature, under a different environment and algorithm.

4.2 A third environment: Level-Based Foraging

LBF (Christianos, Schäfer, and Albrecht 2020) has a public-goods incentive: agents pay an effort cost to load food cooperatively, so the locally rational action is to free-ride. Natively discrete, unrelated to logistics or navigation, trained under centralized-critic MAPPO (matching MPE so only the environment varies), full λ-sweep, same collapse-midpoint criterion (1,600 runs). The risk-neutral null is low and tight ($w_{\text{crit}} \approx 0.08$); curvature moves it *right* by +0.18 (opposite to the supply chain). Loss aversion drives w_{crit} from 0.26 at λ=1 to 0.83 at λ=4, monotone with non-overlapping endpoint CIs (the pre-registered criterion), robust to the collapse threshold. For $\lambda \geq 5$ the boundary saturates against the centralization ceiling and those points are reported as exploratory; an added intermediate level w=0.90 tightens the mid-λ estimates and confirms the saturation.

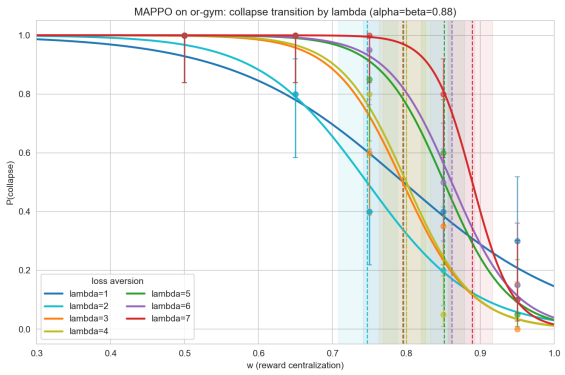


Figure 2: Centralized-critic MAPPO on the supply chain: $P(\text{collapse} \mid w)$ for λ=1, . . . , 7 (700 runs). As under PPO, the family shifts rightward with loss aversion, confirming the effect is not an artifact of independent PPO.

The supply chain's affine law does not transfer: with g/c fixed from λ=1, the best-fit slope saturates at its bound and misfits LBF by RMSE 0.15, converting the form's substrate-specificity from assertion to measured result.

4.3 Cross-algorithm: MAPPO on the supply chain

To isolate the algorithm from the environment, we rerun the full sweep on the *identical* substrate, grid, seeds, label, and hyperparameters, changing only the learner from independent PPO to centralized-critic MAPPO (700 runs; Figure 2). w_{crit} rises $0.796\,[0.742, 0.860]$ at λ=1 to $0.889\,[0.856, 0.917]$ at λ=7, monotone from λ=2 $(0.747, 0.795, 0.800, 0.851, 0.861, 0.889)$, endpoint CIs non-overlapping. The direction generalizes while the absolute level does not: MAPPO sits above PPO at low λ (most visibly λ=1, 0.796 vs 0.586, where MAPPO's transition is shallow and the estimate noisier), converging by $\lambda \geq 4$ ($|\Delta w_{\text{crit}}| \leq 0.04$).

5 Two Boundary-Condition Ablations

5.1 Action representation

The mechanism predicts that destroying the continuous policy gradient removes the boundary. Value-based learners need discrete actions, so we discretized the base-stock control into K=11 bins and ran IQL and QMIX on the identical reward pipeline, w-grid, and label; both produced a flat, w-independent collapse rate with no fittable w_{crit}. To isolate the cause, a single-variable control—PPO on the same K=11 discrete shell—also goes flat (Table 6). Discretization *alone* erases the boundary, so the value-based nulls need no appeal to the learning paradigm. This is not a claim that discreteness per se removes the boundary: MPE is natively discrete and retains it. Coarse bins snap the policy to a stable interior level and never visit the near-threshold (s, S) region where collapse occurs—exactly the continuous-gradient prediction. The effect is a coarse-discretization boundary condition, substrate-specific like the curvature sign, and a confirmed prediction, not a negative result.

Table 6: Action-representation ablation. The control (last row) changes only the action space while holding the learner at PPO, isolating coarse discretization—not the value-based paradigm—as what erases the boundary.

Learner	Action space	Boundary?
PPO / MAPPO	continuous	yes
QMIX (centralized value)	discrete K=11	no
IQL (independent value)	discrete K=11	no
PPO (control)	discrete K=11	**no**

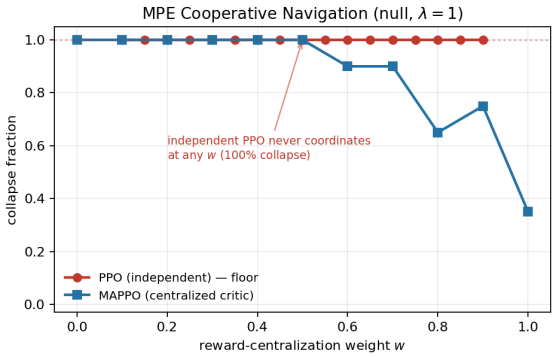


Figure 3: Credit-assignment ablation on MPE (null, λ=1). Independent PPO collapses at every w (a coordination floor), while centralized-critic MAPPO crosses the boundary. The behavioral boundary is exposed only once credit assignment is handled.

5.2 Credit assignment

The boundary is measurable only with a learner that can coordinate on a shared-reward task. On MPE, independent PPO never reaches the coordinated regime at *any* w (collapse fraction 1.0 throughout, including maximal centralization; Figure 3), so behavioral collapse cannot be distinguished from inability to assign credit. Centralized-critic MAPPO recovers coordination (collapse fraction falls to 0.35, occupancy crosses the threshold near $w \approx 0.6$). The same floor appears on LBF. The boundary is therefore measurable only with a centralized critic on shared-reward tasks—why we match one to such environments: a necessity, not a convenience.

6 Pre-registered Signal-to-Noise Probe

The mean-field account calibrates the magnitude of w_{crit} rather than predicting it, because the drift and diffusion coefficients are properties of the stochastic learn-

ing dynamics. We pre-registered a direct measurement to upgrade magnitude from calibrated to predicted, fixing its analysis before running. **Observable:** during PPO training we log the mean base-stock action μ once per update and, over a fixed early window, estimate drift $D(w)$ (mean increment) and diffusion $\Theta(w)$ (increment variance/2), forming $\mathrm{SNR}(w) = D(w)/\sqrt{2\Theta(w)}$. **Decision rule:** sweep $w \in \{0.50, \ldots, 0.90\}$ at the structural condition, 20 seeds/w, 150k steps; the mechanism is corroborated iff the measured crossing's bootstrap 95% CI contains the matched-horizon boundary $w_{\mathrm{crit}} = 0.822$. **Outcome (reported in full):** the collapse-fraction of this independent sweep crosses 0.5 at $w = 0.82$, reproducing the matched-horizon boundary almost exactly—a positive robustness check on the boundary. The *drift-sign* observable, however, was *inconclusive*: the restoring drift in the contested band is weakly positive across all w with no genuine zero-crossing, so it does not localize w_{crit}. The reason is basin geometry—PPO initializes above the cooperative optimum ($S{\approx}250$ vs $S{\approx}150$), and collapses plunge through the contested band too quickly to sample, biasing the conditioned drift toward survivors. We therefore do not claim a magnitude-level prediction; the sign-level and functional-form predictions are unaffected. The driver and analysis (with the decision rule encoded) are released with the code so the test can be re-run with alternative observables.

7 Exploration-Noise Ablation

The mechanism predicts w_{crit} rises with exploration noise. Sweeping the PPO entropy coefficient at the structural condition (300k, 20 seeds, $w \in \{0.5, \ldots, 0.9\}$), the boundary rises monotonically: $w_{\mathrm{crit}} = 0.685$ at ent=0 (reproducing the null), 0.690 at 0.001, 0.715 at 0.01, and beyond 0.05 the system collapses at every tested w (Table 7). **A binding-noise dissociation.** Repeating under MAPPO on the transfer substrates: on LBF the prediction replicates (w_{crit} rises $0.13 \to 0.37$ over ent $\in \{0, 0.05\}$, shift +0.24, non-overlapping endpoints); on MPE the boundary does *not* move (shift +0.05, overlapping CIs; pooled slope over 1,320 runs positive but not significant, β=+0.47, CI $[-3.4, +4.3]$). This is a dissociation predicted by the mechanism: exploration noise moves the boundary only when it is the binding term in the SNR. The supply chain and LBF have sparse, sharp landscapes where injected entropy dominates; MPE's dense distance-shaped reward carries high intrinsic exploration that swamps the added term. The effect appears exactly where the mechanism says it should and is absent where it says it should be—a sharper test than uniform replication.

8 Threshold Robustness

We re-estimate w_{crit} on Exp A @300k under nine variants of the collapse definition, varying the $\bar{S}$ threshold over $\{8, 10, 12\}$ and the $\bar{\pi}$ threshold over $\{-102.5, -127.1, -152.5\}$. Three variants with $\bar{\pi} \leq -152.5$ are degenerate (the minimum profit in the dataset is -128.1) and excluded. Across the remaining six estimable variants the maximum deviation from the baseline $w_{\mathrm{crit}} = 0.666$ is 0.015, well below the CI width: the collapse definition is robust to ±20% threshold perturbation.

Table 7: Entropy-coefficient ablation (α=β=1, λ=1, 300k, 20 seeds/cell). w_{crit} with bootstrap 95% CIs.

entropy coef.	w_{crit} (95% CI)
0.000	0.685 [0.656, 0.710]
0.001	0.690 [0.667, 0.715]
0.010	0.715 [0.689, 0.743]
0.050	> 1 (collapse at all w)

9 Notation and Abbreviations

Symbol	Meaning
w	reward centralization weight, $\in [0, 1]$
w_{crit}	critical w where $P(\mathrm{collapse})$=0.5
λ	loss-aversion coefficient (λ>1)
α, β	CPT value-function curvature (gain/loss)
ρ	loss-domain share of marginal incentive, $\in [0, 1]$
g, c	marginal global benefit / local cost of cooperation
$D(w), \Theta(w)$	Langevin drift / diffusion of the policy mean
$\mathrm{SNR}(w)$	$D(w)/\sqrt{\Theta(w)}$, escape signal-to-noise ratio
σ, η, B	exploration scale / learning rate / batch size
$\Delta U, \tau$	escape barrier / Kramers first-passage time
T	training horizon (environment steps)
$\bar{S}, \bar{\pi}$	mean base-stock / mean profit (collapse label)

Abbreviations. MARL (multi-agent reinforcement learning); CTDE (centralized training, decentralized execution); CPT (cumulative prospect theory); PPO (proximal policy optimization); MAPPO (multi-agent PPO with a centralized critic); MPE (multi-agent particle environments); LBF (Level-Based Foraging); SNR (signal-to-noise ratio); CI (confidence interval).

10 Claim–Evidence Ledger

Table 8 states how broadly each headline claim is supported. The loss-aversion direction is the generalizing core (three environments, two algorithms); the curvature sign and the downstream-minority cascade are supply-chain-specific, the latter framed as a falsifiable prediction for directed-flow domains.

Table 8: Each headline claim and the breadth of evidence behind it. Environments: SC (supply chain), MPE, LBF. Algorithms: PPO, MAPPO.

Claim	Envs	Algos
Sharp collapse boundary w_{crit} exists	SC, MPE, LBF	PPO, MAPPO
Metastability (w_{crit} slides with T)	SC, MPE, LBF	PPO, MAPPO
Loss aversion *raises* w_{crit} (dir.)	SC, MPE, LBF	PPO, MAPPO
Affine λ-law (form)	SC	PPO
Mechanism fixes sign a priori	SC, entropy abl.	PPO
Curvature *lowers* w_{crit} (sign)	SC (opp. on LBF)	PPO
Downstream-minority cascade	SC only	PPO

References

Christianos, F.; Schäfer, L.; and Albrecht, S. 2020. Shared Experience Multi-Agent Reinforcement Learning. In *Advances in Neural Information Processing Systems*, volume 33, 10707–10717.

Gijsbrechts, J.; Boute, R. N.; Van Mieghem, J. A.; and Zhang, D. J. 2022. Can Deep Reinforcement Learning Improve Inventory Management? Performance on Lost Sales, Dual-Sourcing, and Multi-Echelon Problems. *Manufacturing & Service Operations Management*, 24(3): 1349–1368.

Hubbs, C. D.; Perez, H. D.; Sarwar, O.; Sahinidis, N. V.; Grossmann, I. E.; and Wassick, J. M. 2020. OR-Gym: A Reinforcement Learning Library for Operations Research Problems. arXiv:2008.06319.

Leluc, R.; Kadoche, E.; Bertoncello, A.; and Gourvénec, S. 2023. MARLIM: Multi-Agent Reinforcement Learning for Inventory Management. arXiv:2308.01649.

Lowe, R.; Wu, Y.; Tamar, A.; Harb, J.; Abbeel, P.; and Mordatch, I. 2017. Multi-Agent Actor-Critic for Mixed Cooperative-Competitive Environments. *Advances in Neural Information Processing Systems*, 30.

Mengesha, I.; Sun, M.; and Roy, D. 2025. Cognitive Biases Shape the Evolution of Zero-Sum Norms. *arXiv preprint arXiv:2511.16453*.

Oroojlooyjadid, A.; Nazari, M.; Snyder, L. V.; and Takáč, M. 2022. A Deep Q-Network for the Beer Game: Deep Reinforcement Learning for Inventory Optimization. *Manufacturing & Service Operations Management*, 24(1): 285–304.

Raffin, A.; Hill, A.; Gleave, A.; Kanervisto, A.; Ernestus, M.; and Dormann, N. 2021. Stable-Baselines3: Reliable Reinforcement Learning Implementations. *Journal of Machine Learning Research*, 22(268): 1–8.

Schulman, J.; Wolski, F.; Dhariwal, P.; Radford, A.; and Klimov, O. 2017. Proximal Policy Optimization Algorithms. arXiv:1707.06347.

Terry, J. K.; Black, B.; Grammel, N.; Jayakumar, M.; Hari, A.; Sullivan, R.; Santos, L.; Perez, R.; Horsch, C.; Dieffendahl, C.; Williams, N. L.; Lokesh, Y.; and Ravi, P. 2021. PettingZoo: Gym for Multi-Agent Reinforcement Learning. arXiv:2009.14471.

Terry, J. K.; Black, B.; and Hari, A. 2020. SuperSuit: Simple Microwrappers for Reinforcement Learning Environments. arXiv:2008.08932.

Yu, C.; Velu, A.; Vinitsky, E.; Gao, J.; Wang, Y.; Bayen, A.; and Wu, Y. 2022. The Surprising Effectiveness of PPO in Cooperative Multi-Agent Games. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*.